



Interplay between temperature and bandgap energies on the outdoor performance of perovskite/silicon tandem solar cells

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Perovskite/silicon tandem solar cells promise power conversion efficiencies beyond the Shockley-Queisser limit of single-junction devices; however, their actual outdoor performance is yet to be investigated. Here we fabricate 25% efficient two-terminal monolithic perovskite/silicon tandem solar cells and test them outdoors in a hot and sunny climate. We find that the temperature dependence of both the silicon and perovskite bandgaps—which follow opposing trends—shifts the devices away from current matching for two-terminal tandems that are optimized at standard test conditions. Consequently, we argue that the optimal perovskite bandgap energy at standard test conditions is <1.68 eV for field performance at operational temperatures greater than 55 °C, which is lower compared with earlier findings. This implies that bromide-lean perovskites with narrower bandgaps at standard test conditions—and therefore better phase stability—hold great promise for the commercialization of perovskite/silicon tandem solar cells.

Metal halide perovskites have attracted considerable attention for their application in tandem solar cells thanks to their high absorption coefficient, small Urbach energy (E_U)^{1,2} and the tunability of their bandgap energy (E_g)³. Perovskite/silicon tandem solar cells are particularly appealing as they pair proven high power conversion efficiencies (PCEs)^{4–9} (values >29% have already been reported¹⁰) with the industrial maturity of crystalline silicon (c-Si) photovoltaics (PVs)^{11–13}. Furthermore, unlike III–V/c-Si tandems, metal halide perovskites can be deposited by facile, versatile and potentially low-cost methods, even on macroscale textured wafers, which are ubiquitous in c-Si PV manufacturing^{5,8,14–16}.

The radiative efficiency limit of perovskite/silicon tandems is ~44% under standard test conditions (STCs, that is, AM1.5G spectrum, 1,000 W m⁻², 25 °C); however, for monolithic, two-terminal (2T) tandems, this requires perfect current matching between the perovskite top cell and the c-Si bottom cell. For this, the perovskite E_g should be ~1.73 eV at STCs (Supplementary Fig. 1a)^{11,17,18}, which is significantly larger than the bandgaps typically used for leading single-junction perovskite devices (~1.50–1.60 eV)^{19–21}. Conventionally, to open the E_g of halide perovskites, the Br:I⁻ ratio in the precursor (and thus the crystallized film) is increased²². However, beyond a Br:I⁻ ratio of ~1:4 (corresponding to an E_g of ~1.68 eV), iodide- and bromide-rich domains segregate on illumination, degrading device performance^{23–25}. The E_g of semiconductors also generally depends on temperature: as temperature increases, E_g narrows for c-Si but increases for perovskites^{1,26}. This is relevant for PVs as outdoor field-testing conditions (FTCs) often imply elevated operating temperatures of PV modules well above 25 °C, especially for high solar irradiation and space-constrained rooftop applications (due to heat-trapping effects)²⁷. Thus, what may be an ideal perovskite E_g for tandems at STCs is probably not ideal under actual FTCs. First, due to the red-shift of the c-Si E_g with

temperature, the required perovskite E_g for current matching will already be lower than 1.73 eV. Second, due to the blue-shift of the perovskite E_g with temperature, its optimal E_g measured at STCs will be even lower than that at FTCs, impacting the design guidelines for efficient tandems.

In this work we experimentally explore the impact of temperature-dependent changes to the E_g of perovskites and c-Si on 2T tandem performance. We analysed tandems with a certified PCE of 25% using double-side-textured silicon heterojunction (SHJ) bottom cells coated with perovskite top cells. We first experimentally measured the tandem temperature under actual outdoor conditions by using field-test facilities at KAUST. We then calibrated a device-temperature model with these data (details are given in the Methods). By combining this model with relevant meteorological data, we simulated the tandem temperature over a complete year (2016), giving a realistic framework for tandem-performance analysis. Second, we experimentally characterized our tandems in a controlled laboratory environment under various temperatures and illumination spectra, with specific attention paid to their impact on current matching. Finally, to gain further insight we developed a temperature-dependent radiative efficiency model to support our findings. This modelling reveals that the optimal perovskite E_g for operation under FTCs greater than 55 °C is <1.68 eV (when measured under STCs). Notably, such narrower- E_g perovskites usually feature higher chemical stability^{24,28}. Overall, these findings provide a clear pathway towards commercial perovskite/silicon tandems that combine high performance with high stability^{29–31}.

Double-side-textured monolithic perovskite/silicon tandems

We fabricated monolithic 2T tandems on double-side random-pyramid-textured c-Si wafers. Such texturing effectively

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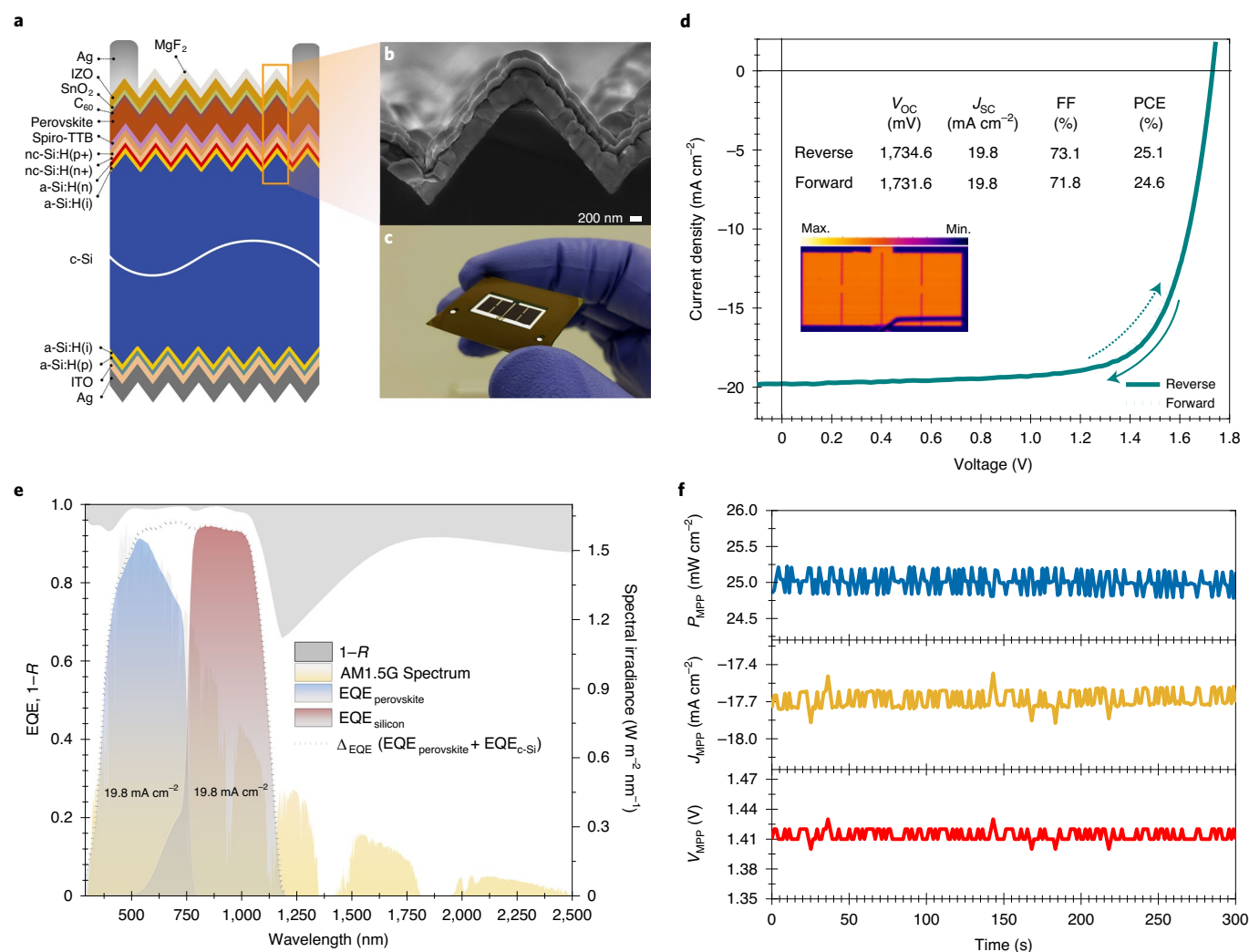


Fig. 1 | Design and performance of the perovskite/silicon tandem solar cells. a–c, A schematic illustration (a), scanning electron microscopy cross-sectional view (b) and a photograph (c) of the double-side-textured tandem solar cells prepared in this work with a hybrid-processed perovskite absorber layer. The detailed contact layout is given in Supplementary Fig. 5. **d,** J - V characteristics of the certified champion device, measured at Fraunhofer ISE CalLab. The inset shows a light-beam-induced current image under 532 nm excitation, demonstrating the uniformity of the perovskite top cell. The colour bar shows the signal intensity. The perovskite absorber layer has an E_g of 1.63 eV. **e,** The corresponding EQE and 1- R data for the same device shown in d at 25 °C overlaid on the AM1.5G solar spectral irradiance. The given J_{SC} values represent the integrated values from EQE measurements of each subcell. **f,** Photovoltaic parameters of a perovskite/silicon tandem under MPP tracking conditions for 5 min. P_{MPP} , power at MPP; J_{MPP} , current density at MPP; V_{MPP} , voltage at MPP.

couples light into c-Si and is obtained with established industrial processes. Silicon heterojunction technology was then used to fabricate the c-Si bottom cell, which was subsequently conformally coated by the perovskite top cell using the hybrid-deposition technique⁵. Figure 1a–c depicts the geometry, cross-sections and a photograph of the device. The top and bottom cells are electrically interconnected via a nanocrystalline silicon recombination junction, which combines a low tunnel resistance with a high shunt resilience and adequate coupling of near-infrared light into the c-Si bottom cell³².

Figure 1d shows the current density–voltage (J - V) characteristics of our champion device with a certified PCE of 25.1% (steady-state maximum power point (MPP) of 25% over 0.8317 cm^2 aperture area, see Supplementary Fig. 2). These results are consistent with our in-house J - V measurements (Supplementary Fig. 3). The relatively narrow statistical distribution of the device parameters among independent tandem batches attests to the reproducibility of the subcell and tandem processing (Supplementary Fig. 4). We studied the spectral response of the tandems by external quantum efficiency (EQE) and 1- R measurements (R is reflectance, Fig. 1e). The tandems

show <2% average reflection in the range of 290–1,100 nm, which is instrumental³³ in enabling a short-circuit current density (J_{SC}) of 19.8 mA cm^{-2} , one of the highest values in the literature (Supplementary Table 1). Between 1,200–2,500 nm, we find the average front-escape reflection (photons internally reflected off the rear SHJ c-Si contact and not absorbed) to be only 15% in our tandems; all other photons in this spectral range are absorbed in the opaque rear contact or by free carrier absorption in the front transparent conductive oxide (TCO), contributing to device heating³⁴.

Perovskite/silicon tandems under outdoor field conditions

We first evaluate the tandem performance under direct sunlight and environmental factors, which is increasingly advocated for as perovskite-based technologies progress towards market readiness³⁵. The devices were vacuum laminated between glass and a transparent back sheet with polyolefin as an encapsulant to protect the devices from environmental contaminants such as oxygen and water (see Methods and Supplementary Fig. 6a,b for details). As expected, the field-measured J_{SC} of the tandem follows the solar irradiation

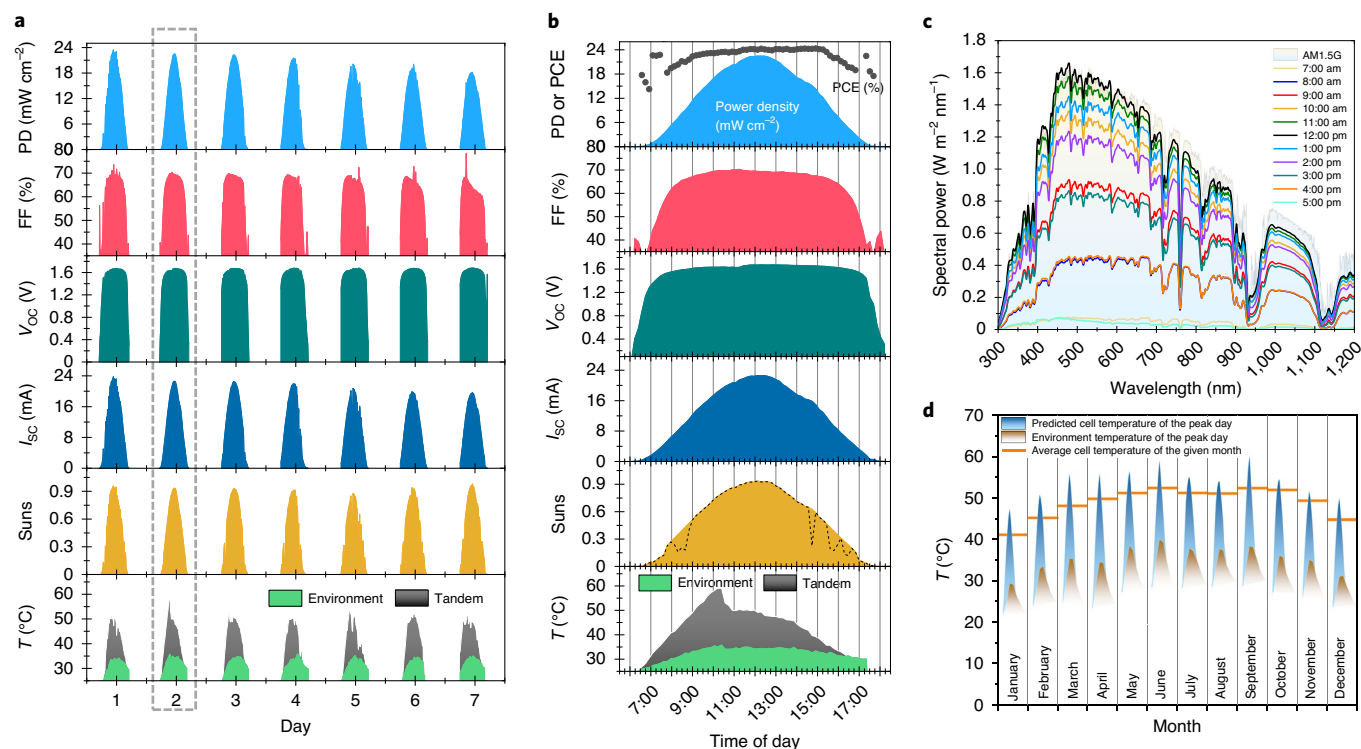


Fig. 2 | Outdoor performance of perovskite/silicon tandem solar cells. **a**, Photovoltaic parameters of a representative perovskite/silicon tandem solar cell tested outdoors at KAUST during a week in November 2019, the corresponding solar irradiance as well as ambient and dummy device temperatures (active area of 50 cm^2 without a front electrode). Note that the I_{sc} of the tandem device is calculated based on the aperture opening on the encapsulation glass, which was 1.17 cm^2 . PD, power density. The decrease of the FF values are assigned to contact degradation as discussed in Supplementary Note 3. **b**, A magnification of the data highlighted in **a** (day 2). The dashed line in the Suns data represents the measured Suns values. The orange shaded values represent the interpolated data, free of shading. The intensity fluctuations in the measured Suns data are due to local shading on the pyranometer. The sharp decrease in device temperature at $\sim 11:00 \text{ am}$ is due to an uptake of the wind speed (as shown in Supplementary Note 4), resulting in a slight PCE increase, underlining the effectiveness of convective cooling. The PCE values were calculated by dividing the power density by the Suns data. **c**, Spectral power over one day measured at KAUST on November 7 2019, along with the Suns data in **b**. **d**, Ambient temperature and predicted cell temperature values for a perovskite/silicon tandem device over a complete year (based on weather data of 2016, KAUST). The tabulated data for **d** are given in Supplementary Fig. 18b. The day in each month with the highest peak temperature values is displayed. The orange lines represent the average peak temperature of the month. The given temperature values in **d** represent the 24 h timeframe for each day.

(Fig. 2a). We find that I_{sc} drops by approximately 0.5 mA cm^{-2} under 1 Sun illumination when devices are encapsulated because of front-glass reflection (Supplementary Fig. 7). We also note that the solar spectrum measured at KAUST is relatively weak in the infrared compared with the standard AM1.5G spectrum, as shown in Fig. 2c and Supplementary Fig. 8b. This will affect current matching for tandems that are optimized for STCs, as detailed in Supplementary Note 1 (refs. 29,36). The V_{oc} and fill factor (FF) values reached their maximum at $\sim 9:30 \text{ am}$ (at about 0.5 Sun) and maintain these values until $\sim 4:30 \text{ pm}$. In the field, the maximum V_{oc} values are lower than those measured in the laboratory (Fig. 2b), which is partially due to the lower light intensity, but especially due to the device temperature (here T_{cell} is $\sim 50^{\circ}\text{C}$ and $\sim 60^{\circ}\text{C}$ for encapsulated and unencapsulated devices, respectively, see Supplementary Note 2 for details). A high operating temperature is generally undesirable for solar cells as it reduces the PCE; the extent of which is expressed by the temperature coefficient (TC, defined as the change in PCE per temperature increment, normalized to the PCE at STCs). Overall, we find that over the course of a given day, the energy yield of the tandem solar cell is mostly dictated by its I_{sc} . In this study, we therefore specifically explore the impact of temperature on the I_{sc} of 2T tandem solar cells.

To explore the impact of temperature, we first estimated the expected T_{cell} range under realistic operation conditions. For this, we fabricated 50 cm^2 dummy tandems (without front metallization)

with a thermocouple attached to their rear, followed by vacuum lamination between two glass sheets with a butyl rubber edge sealant (Supplementary Fig. 6b,d). These samples were exposed (under open-circuit conditions) to direct sunlight for a week in November 2019, at KAUST. For an ambient temperature ($T_{amb,max}$) of 38.4°C , $T_{cell,max}$ reached as high as 60°C (Fig. 2d). For these outdoor measurements, we used an open-rack test configuration (Supplementary Fig. 6c); considerably higher T_{cell} values can be expected in summer and for rooftop installations because of heat-trap effects³⁷. Moreover, the device area here is much smaller than the encapsulation glass. We therefore expect that T_{cell} in commercial-scale modules will be even higher than the reported values here due to their substantially higher device-packing densities.

These data, combined with local meteorological data, were then used as inputs to a thermal model (detailed in the Methods), yielding simulated tandem temperatures over a complete year. Based on these calculations, a maximum perovskite/silicon T_{cell} of 60°C was predicted; $\sim 75^{\circ}\text{C}$ therefore seems to be a realistic temperature for certain rooftop applications and/or under high solar irradiation^{27,38}.

Temperature-dependent characteristics of the tandem cells

Next, we performed temperature-dependent $J-V$ measurements in a controlled laboratory environment under a simulated AM1.5G

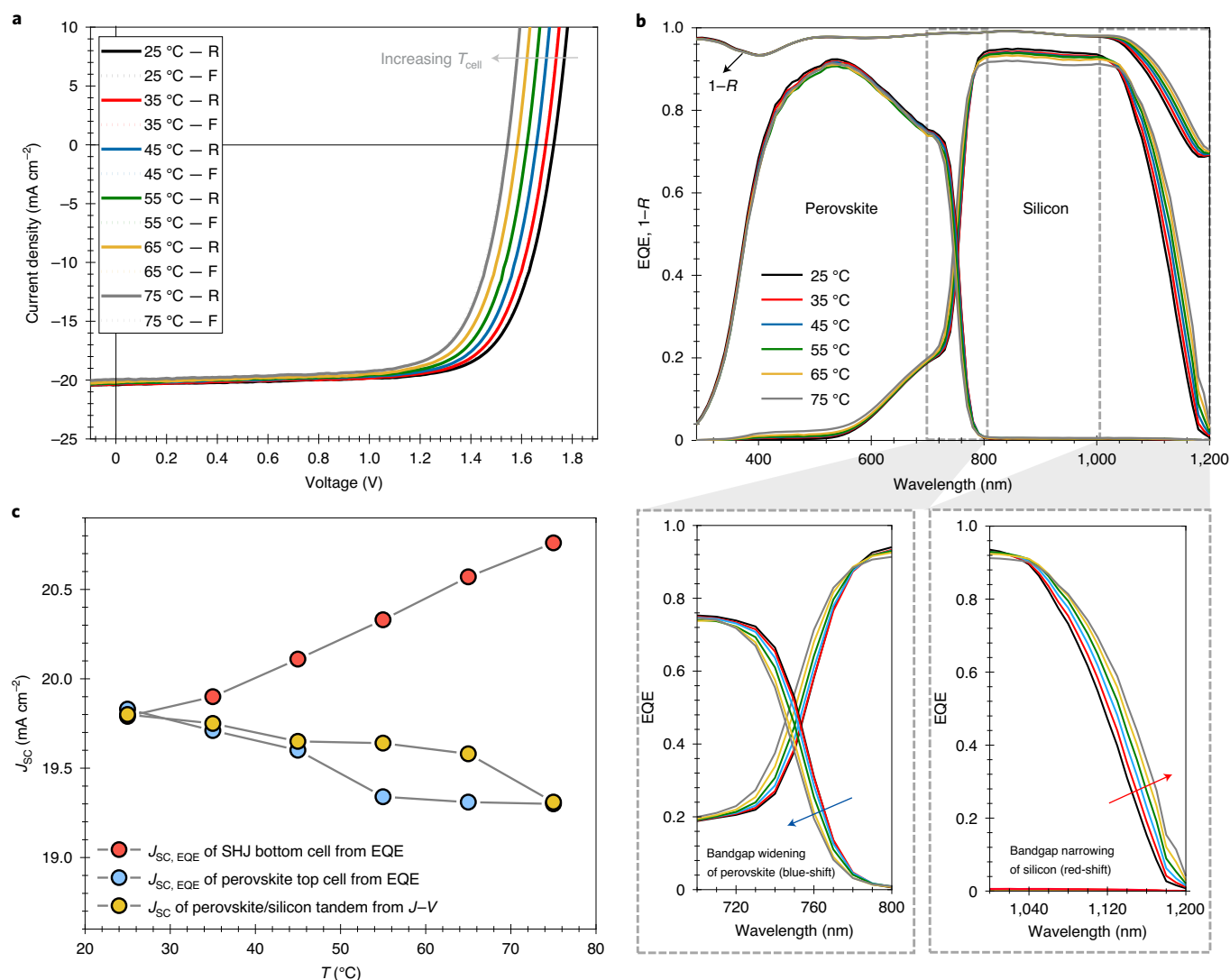


Fig. 3 | Temperature-dependent performance of the perovskite/silicon tandem solar cells. a, b, Temperature-dependent J - V (R, reverse scan; F, forwards scan) (a) and EQE and $1-R$ (R is reflectance) characteristics (b) of perovskite/silicon tandem devices. The bottom of b shows a magnification into the region of the band edge for both subcells. c, Temperature-dependent change of EQE measured current density ($J_{SC,EQE}$) values for both subcells together with the J - V measured current density (J_{SC}) of a tandem solar cell, showing current limiting conditions.

spectrum, within the expected operational temperature range, by varying the temperature between 25 and 75 °C (Fig. 3a). From these data, we calculated the TC for our tandem to be $-0.26\% \text{ K}^{-1}$ (see Supplementary Note 5)³⁹. The c-Si bottom cell used in this study showed a TC of $-0.23\% \text{ K}^{-1}$ when normalized at 25 °C; a TC of $-0.26\% \text{ K}^{-1}$ —which is more typical for SHJ technology—was calculated when normalized at 40 °C (which is where the FF is linear with temperature; see Supplementary Fig. 20)³⁹. The addition of the perovskite top cell therefore barely affects the tandem TC. Notably, aside from the physical factors that dictate the TC of single-junction solar cells⁴⁰, the TC of 2T tandem solar cells is also affected by the possible current mismatch between the top and bottom cells. In the tandem under study here, we find this increasingly to be the case with temperature, resulting in a decreased J_{SC} of the device, thereby inflating the TC of the tandem. This current mismatch is due to the E_g shifts of c-Si and perovskite layers with temperature: the c-Si E_g narrows as temperature increases, whereas the perovskite E_g broadens. This implies that the TC of tandems can be minimized by optimizing the perovskite E_g towards current matching at the highest expected operational temperature. This possible mismatch-induced

increase of the tandem TC is also of importance when comparing TCs of tandems as well as their PCEs at STCs in the literature.

To further elucidate the E_g shift as a function of temperature, we studied the temperature-dependent EQE of the tandems (Fig. 3b). Figure 3c shows that the $J_{SC,EQE}$ (obtained by spectrally integrating the product of EQE and incident solar spectrum) of the c-Si bottom cell increases with temperature, which is due to additional contributions of photons with longer wavelengths (as the c-Si E_g red-shifts) as well as photons with shorter wavelengths (as the perovskite E_g blue shifts), coupling more light from two distinct spectral ranges in the c-Si cell underneath. Despite this, the overall J_{SC} of the 2T tandem decreases because the $J_{SC,EQE}$ of the perovskite top cell decreases (as its E_g blue-shifts), eventually limiting the overall J_{SC} ; thus, what may be the optimal perovskite E_g for STC (that is, at 25 °C) may not be optimal for outdoor applications, where T_{cell} is often considerably higher.

Impact of spectral mismatch on tandem performance

To gain further insight into the current mismatch in our tandem cells due to variations in both the solar spectrum and FTC operating temperature, we intentionally modified the incident spectrum of our

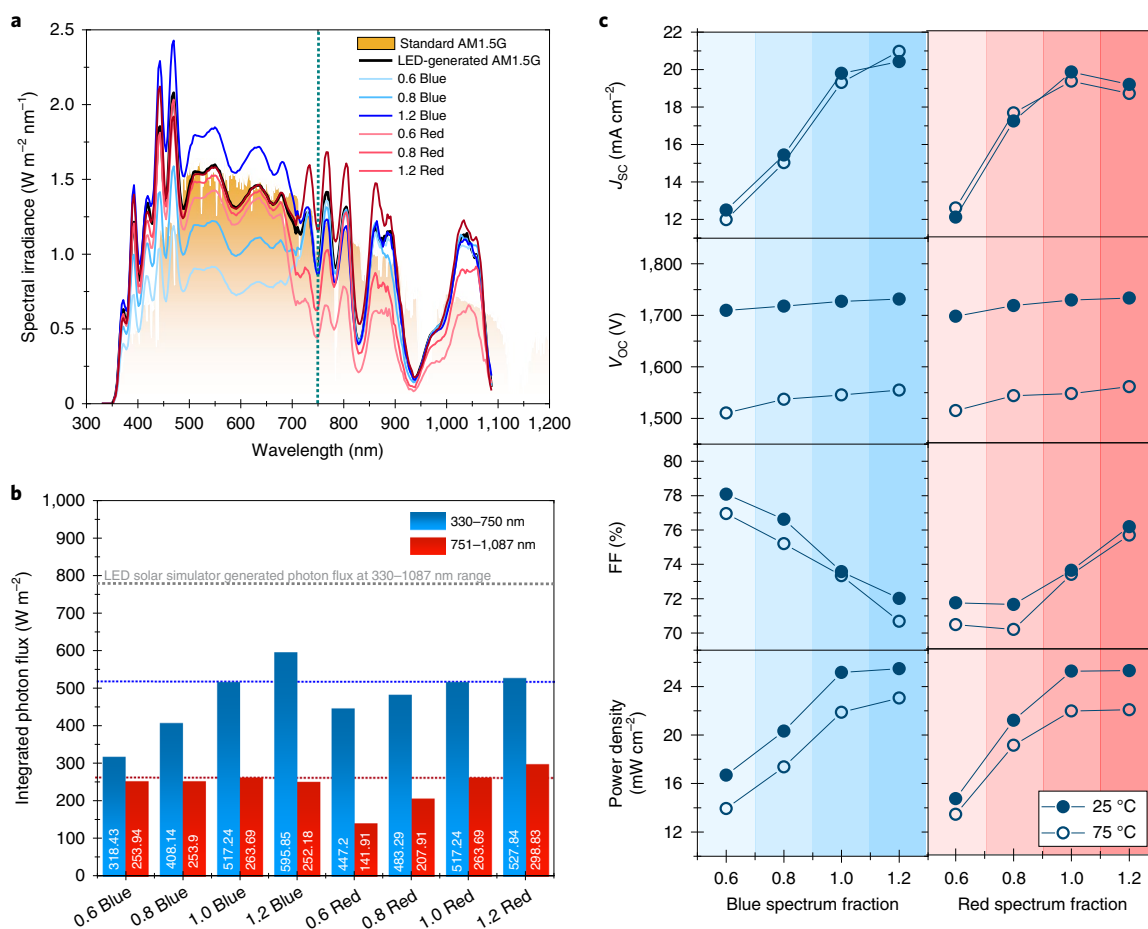


Fig. 4 | The performance of the tandem solar cells under current mismatch conditions. a, b, The generated spectrum of the different blue and red regions together with standard AM1.5G spectra (**a**) and corresponding integrated photon flux values for the spectrum generated via an LED-based solar simulator (**b**). Here, 1,000 W m^{-2} denotes the standard AM1.5G spectrum; red and blue dashed lines in **b** are guides to the eye that show the blue and red portion of the photon flux for the spectral ratio of 1 generated by the LED. **c,** The change of the device characteristics for various blue and red spectrum fractions for 25 and 75 °C. The intensity of the spectrum fractions is shown in **b** for each red and blue combination. Here, spectrum fraction defines the intensity of the spectrum (for the given range) normalized to the intensity of LED-generated AM1.5G standard spectrum; 75 °C is chosen as a representative of elevated temperatures.

solar simulator relative to the standard AM1.5G spectrum for device measurements at 25 and 75 °C. In this way we mimic the effect of altering the perovskite E_g on the tandem J_{sc} , which is a simple, yet powerful method to investigate current mismatch in 2T tandems⁴¹. Specifically, we altered the AM1.5G spectrum in our light-emitting diode solar simulator by consecutively varying the blue (330–750 nm) and red (751–1,087 nm) parts of the spectrum, as shown in Fig. 4a,b. We find that the J_{sc} of the tandem follows the same trend regardless of whether these blue or red parts were decreased, as illustrated in Fig. 4c; however, increasing these respective spectra above AM1.5G impacts the J_{sc} in opposite directions. Increasing the blue spectral part increased the J_{sc} , which suggests that our tandem is slightly current mismatched at STCs, caused by a current-limiting perovskite top cell. When—in addition to these spectral modifications— T_{cell} was increased to 75 °C, the relative increase in J_{sc} was more extensive, as now the c-Si subcell $J_{sc,EQE}$ is also increased (due to the temperature-induced perovskite and c-Si E_g changes, as discussed above). Conversely, increasing the red spectral contribution slightly decreased the J_{sc} of the device. We also find that the current-dominating subcell always dictates the value of the tandem FF, as shown in Fig. 4c, which is in line with previous findings⁴¹.

To further maximize the tandem J_{sc} , internal light trapping in the c-Si cell should be improved beyond the use of textured c-Si wafers,

for example, by suppression of parasitic infrared light absorption in its rear contact stack, as advocated for single-junction SHJ devices³³; the perovskite E_g should then be adapted accordingly. The temperature of the tandem cell should also be considered when determining the optimal perovskite E_g to enable current matching and maximize the J_{sc} at FTCs.

Temperature-corrected bandgap of the perovskite top cells

To further elucidate the effect of temperature on current matching in perovskite/silicon 2T tandem solar cells, and provide tandem design guidelines with respect to the perovskite E_g , we modelled their temperature-dependent radiative efficiency limit on the basis of the modelling methodology outlined by Jean and colleagues⁴². Our adapted model considers the effect of the temperature-dependent perovskite E_g and its E_U on tandem performance; further details are given in the Methods. As discussed above, in the case of 2T tandems, a departure from current-matching conditions leads to a substantial drop in the maximum PCE. In Supplementary Fig. 1a,b, the traditional Shockley–Queisser efficiency limit (which only considers E_g values) for 4T and 2T perovskite/silicon tandems was calculated at 25 °C and is plotted as a function of the perovskite E_g . In the 2T case, a departure from the optimal, current-matched perovskite E_g of 1.73 eV results in tandem power losses, either due to the perovskite or c-Si

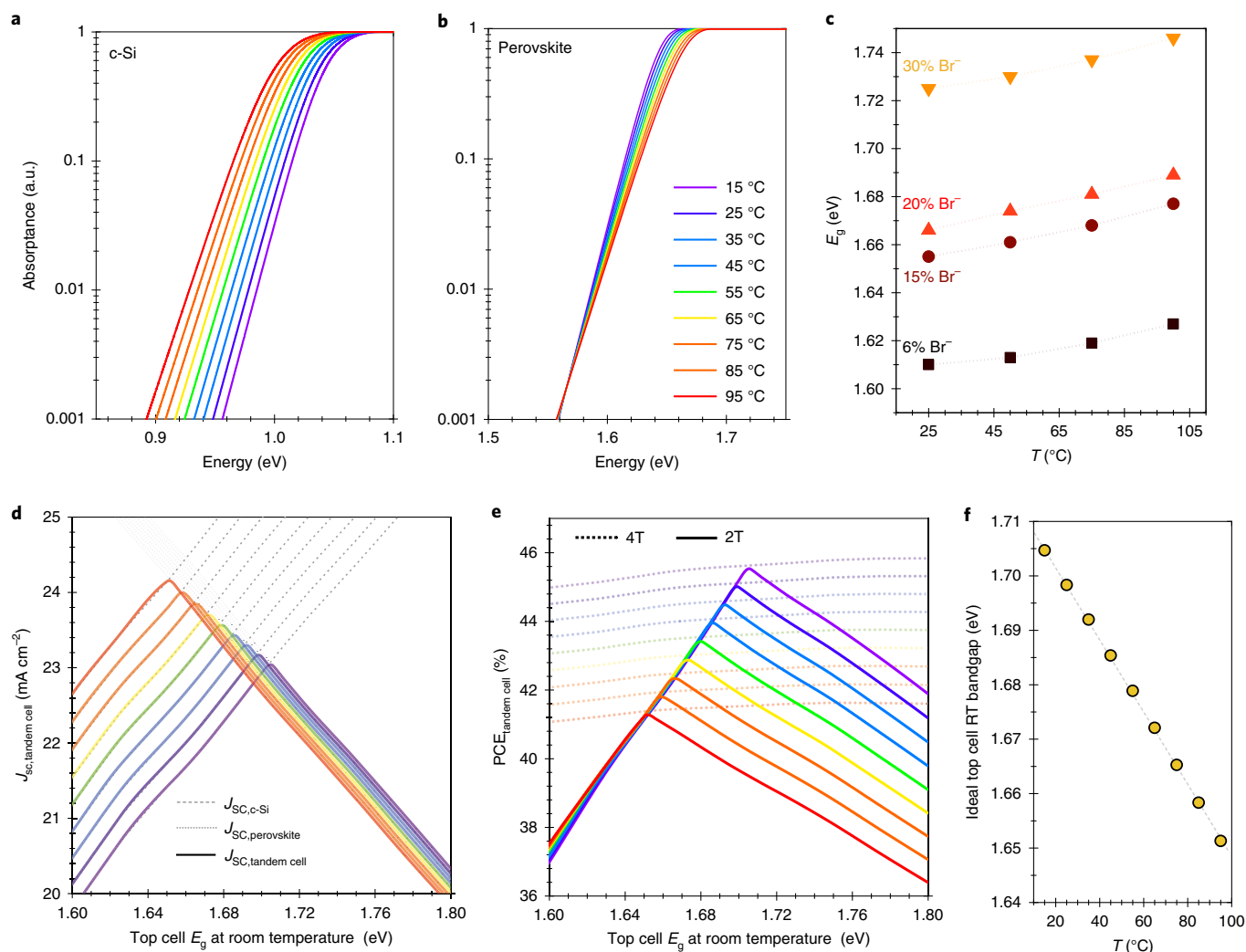


Fig. 5 | The calculated temperature-dependent performance of the ideal perovskite/silicon tandem solar cells. a–c, The simulated band edge absorbance for c-Si (**a**) and perovskite semiconductors (**b**) considering the temperature dependencies of E_U and E_g . Note that the simulated perovskite absorbance data represents a room temperature E_g of 1.66 eV and that the slope of the E_g change was 0.31 meV K^{-1} , as calculated from the experimentally acquired ultraviolet–visible–near infrared absorbance of 20% bromide content (**c**) in the mixed-halide perovskite films deposited on glass substrates as shown in Supplementary Fig. 22. The slope of the E_g change for silicon was $-0.273 \text{ meV K}^{-1}$. The dotted lines are guides to the eye. **d,** The variation of the total 2T tandem cell current as a function of the top cell perovskite E_g and cell temperature. The dotted lines show the current contribution from the perovskite subcell, whereas the dashed lines show the contribution from the c-Si subcell. The maximum efficiency occurs when the two values are equal. **e,** The corresponding PCE values for both 2T (solid lines) and 4T (dotted lines) configurations as a function of top cell E_g and temperature. **f,** The calculated ideal room temperature (RT) E_g of the top cell perovskite absorber at different operating temperatures. The dotted lines are guides to the eye. The temperature legend in **b** also applies to panels **a**, **d** and **e**.

subcell limiting the J_{SC} (for a perovskite E_g lower or higher than 1.73 eV, respectively). This is not the case for the 4T configuration as the subcells operate independently and their output powers sum.

In general, the addition of band tailing to a semiconductor (caused by the structural and thermal disorder, expressed by E_U)² tends to lower its corresponding efficiency limit by increasing the dark recombination current and hence decreasing the device voltage. The voltage deficit ($E_g/q - V_{OC}$, where q is the elementary charge) in solar cells of different materials is known to show a strong correlation with E_U (ref. ¹). Disorder—and thus E_U —scales with temperature; it was recently established that such changes correlate with temperature-induced losses in V_{OC} (ref. ¹). Overall, it is the increase in the dark recombination current with temperature—to which band tailing is a contributor—that drives the absolute decrease in PCE values of both the 4T and 2T device configurations with increasing temperature.

Adding the effect of temperature to both E_g and E_U leads to a clear shift in the peak value of the optimal top cell E_g for 2T tandems. This can be ascribed to the change in E_g of the two materials, the result of which is shown in the absorbance values used in the efficiency calculation shown in Fig. 5a,b. Note that we have assumed composition-independent values for both the perovskite's temperature-dependent $E_U(T)$ and $E_g(T)$ values. In the case of the E_g , our temperature-dependent measurements on a variety of compositions are shown in Fig. 5c (in which the Br:I ratio is altered). The data shows the similar dependency of E_g over the temperature range investigated for all of the perovskite compositions, justifying our assumption in the model^{1,2}.

For the 2T case, we have also plotted the tandem J_{SC} as a function of the top cell E_g and include the current contributions of the individual subcells (dashed and dotted lines in the Fig. 5d). For any

given temperature, the left side (lower values of perovskite E_g energies) of the peak in current denotes the values of the perovskite E_g that induce a constraint in the overall J_{sc} by the c-Si bottom cell. Here the c-Si subcell is shaded by the top cell and hence the tandem J_{sc} follows the subcell current and its temperature dependence. On the right side of the apex, the wide- E_g perovskite current is constraining the device. The inversion in the trend of temperature at large perovskite E_g is due to the asymmetry in the trend of E_g with temperature, as also seen in the EQE(T) spectra of Fig. 3b. When the tandem is current limited by the bottom cell, the increase in temperature induces an increase in infrared absorption and so an increase in the overall J_{sc} . The opposite is true when the device is constrained by the top cell: increasing the temperature results in lower currents in the top cell due to further broadening of the E_g away from the current-matching condition (Fig. 5d), as also demonstrated experimentally. By adding the effect of band tailing and light trapping to the radiative efficiency limit calculations, we show that the ideal perovskite E_g value decreases substantially from the traditional calculation and that this value decreases further with increasing temperature (Fig. 5e). Critically, the optimal perovskite E_g value at the radiative limit decreases to below 1.68 eV for relevant device operating temperatures. By applying the model to the temperature and intensity data of Fig. 2b, we calculate the top cell bandgap that delivers the highest energy yield at the radiative limit to be ~1.685 eV (see Supplementary Fig. 9). Such lower bandgaps are typically associated with materials with modest Br:I ratios, which are known to be more stable. In the case where the optimal top cell bandgap exceeds 1.68 eV, the relative power loss ceded by utilizing a perovskite of 1.68 eV is probably minimal compared with previous determinations of the optimal efficiency, minimizing the incentive to explore wider bandgaps and so less stable perovskite compositions. Of course, these optimizations are sensitive to the input spectrum and will also be affected by parasitic losses in real cells and modules, as demonstrated in Supplementary Fig. 11. Note that, for clarity, we display the data as a function of the top cell E_g at room temperature, although this value is also changing with temperature (Fig. 5f).

Conclusions

We elucidated the temperature dependence of perovskite/silicon tandem solar cells, with specific attention to elevated temperatures typical of an actual device's operating conditions. To do so, we fabricated double-side-textured monolithic perovskite/silicon tandem solar cells with a certified PCE of 25.1% and the highest J_{sc} value reported so far for such devices. We then studied the interplay between the operating temperature and the bandgaps of the subcells and its effect on tandem performance. We tested these devices under both controlled laboratory conditions and outdoor field conditions. We find that, in a hot and sunny climate, perovskite/silicon tandem cells can easily reach ~60 °C under direct sunlight and calculated that this can be as high as ~75 °C over the course of a year during operation.

These observations set a framework for the optimal design of perovskite/silicon-based tandem cells towards maximized power output under actual operating conditions. With increasing temperature, the E_g of c-Si narrows, whereas for perovskites, it broadens, which affects current-matching conditions. Therefore, for optimal tandem performance at field-testing conditions, we find that the perovskite E_g at STCs (25 °C) are lower than earlier findings. This conclusion is further supported by calculations using a modified temperature-dependent radiative efficiency limit model. By applying this model, we find that the optimal perovskite E_g falls below 1.68 eV (measured at 25 °C) at the radiative limit at elevated temperatures, which is consistent with operating temperatures in a hot and sunny climate. This implies that conventional perovskites with moderate Br:I ratios, which do not suffer from halide segregation,

can be used for tandems deployed in hot and sunny climates. Our results overall give a straightforward perovskite-design framework for the development of stable and high-efficiency perovskite/silicon tandems for commercial deployment.

Methods

Fabrication of the silicon bottom cells. Silicon heterojunction bottom cells are fabricated on float-zone double-side-textured 4-inch wafers (n-doped; 1–5 Ω cm resistivity, 250–280 μ m thickness). The texturing process was performed in an alkaline solution to achieve randomly distributed pyramids; the wafers were subsequently cleaned in RCA-1 and RCA-2 solutions. Before the plasma enhanced chemical vapour depositions, the wafers were dipped in a 5% hydrofluoric acid solution to remove the native oxide layer. The intrinsic, n and p amorphous silicon layers were deposited in a plasma enhanced chemical vapour deposition cluster (Indeotec Octopus II) with silane, hydrogen, phosphine and trimethylboron as input gases. The back electrode was formed by sequentially sputtering ITO and Ag (150 and 250 nm, respectively) into the physical vapour deposition part of the Octopus cluster; nc-Si:H(n)/nc-Si:H(p) recombination junction is formed by the sequential deposition of the n- and p-doped silicon films utilizing high hydrogen:silicon gas flow ratios. All depositions were performed at temperatures of ≤ 200 °C.

Fabrication of the perovskite top cells. To achieve p-i-n configuration perovskite top cells, 12 nm of spiro-TTB (>99%, Lumtec) was deposited by Ångström EvoVac thermal evaporation system (base pressure of $<1 \times 10^{-6}$ Torr, evaporation rate of 0.25 Å s⁻¹, quartz crucible) as the hole transport layer on top of the nc-Si:H(n)/nc-Si:H(p) recombination junction. On top, as the first step of the sequential method, PbI₂ (>99.99%, beads from Alfa Aesar) and CsBr (>99.9%, Sigma-Aldrich) were co-evaporated using the same evaporation system ($<1 \times 10^{-6}$ Torr base pressure, evaporation rate of 1 Å s⁻¹ for PbI₂ and 0.1 Å s⁻¹ for CsBr, aluminium oxide crucibles). During the evaporation processes, no intentional heating or cooling was applied to the substrates. Later, as the second step of the process, a precursor of formamidinium iodide and bromide (in 3:1 molar ratio, 0.6 M in ethanol, Greatcell Solar) was spin-casted onto the co-evaporated layer in a nitrogen glove box (spin speed of 4,000 r.p.m. during 35 s, quenching after 5 s). The resultant intermediate phase was then annealed at 150 °C for 30 min in a controlled humidity environment (30%) to form the perovskite absorber. 20 nm C₆₀ was evaporated within the same system ($<1 \times 10^{-6}$ Torr base pressure, evaporation rate of 0.2 Å s⁻¹, quartz crucible). SnO₂ (20 nm) was deposited by the atomic layer deposition technique, using a Picosun system at 100 °C substrate temperature and tetrakis(dimethylamino)tin (TDMASn) and H₂O as precursors (kept in 80 °C, 134 cycles). To form the transparent electrodes, 100 nm of indium zinc oxide was sputtered in an Ångström Engineering system using a 3-inch target (90% In₂O₃/10% ZnO) with a radio frequency power of 42 W ($<1 \times 10^{-6}$ Torr base pressure). The four-point probe measured sheet resistance of the indium zinc oxide films was 40 Ω per square (for 100 nm, on glass). Silver metal fingers (200 nm) were thermally evaporated through the shadow masks with a rate of 2.5 Å s⁻¹. As a final step, 100 nm of MgF₂ antireflection films were thermally evaporated within the same system (random-sized crystals from Plasmaterials, >99.9%, 1 Å s⁻¹, $<1 \times 10^{-6}$ Torr base pressure). All thickness values reported here represent the thickness values on flat substrates; however, for the thermal evaporation onto the textured interfaces, a 1.7 geometric factor ratio (1.7-times thicker) is applied during the deposition.

Fabrication of the various bandgap of perovskite films with the one-step process. PbI₂ and PbBr₂ (Alfa Aesar), methylammonium bromide (Greatcell Solar), formamidinium iodide (Greatcell Solar) and caesium iodide (Sigma-Aldrich) were dissolved in a mixture of DMF:DMSO (4:1, Sigma-Aldrich) to prepare the perovskite precursor solution. Thin films of perovskites (FACsMAPbI_{3-x}Br_x) with different bromide content (6%, 15%, 20%, 30% of the total halide content) were deposited in nitrogen atmosphere via spin coating at two-step spin process (first 2,000 and later 5,000 r.p.m.) on soda-lime glass substrates, using chlorobenzene as a solvent quencher. The films were subsequently annealed on a hotplate at 100 °C for 10 min.

Temperature-dependent bandgap measurements. Carry 5000 (Agilent Technologies) ultraviolet-visible tool was used to measure the absorbance of the fabricated perovskite films. The band gap values of the perovskites with different bromide content at different temperatures. The temperature control was provided via a Janis cryostat (with quartz window) with Peltier heaters and under a nitrogen flushing.

Characterization of the photovoltaic performances. Room temperature and temperature-dependent J - V measurements were performed using Wavelabs Sinus 220 LED-based solar simulator with AM 1.5G irradiance spectrum. The light intensity was calibrated using Fraunhofer ISE CalLab certified c-Si solar cells. The temperature is controlled during the measurement using a vacuum stage using a Peltier element.

Standard $J-V$ measurements were performed at 25 °C. The illumination area of the devices was 1.03 cm², which is determined by the laser cut shadow mask coated with matte black paint. The device measurements were performed at 200 mV s⁻¹ scan speed using Keithley 2400 source meter. The stabilized power measurements (at 25 °C) were recorded via the homemade LabView-based software⁴⁵ using a three-point perturbation method. Room-temperature and temperature-dependent EQE measurements were performed using PV-Tools LOANA equipment by measuring the temperature from the backside of the wafers. When measuring perovskite top cells, the tandem devices were light-biased by infrared light-emitting diodes (930 nm); when measuring c-Si bottom cells, the tandem devices were light-biased by a blue LED (450 nm) to saturate the subcells. To enable the near short-circuit conditions for each subcell, MPP voltages were applied on the devices. For in-house measurements, all device characterizations were performed at ambient air with average RH ~50% and without any encapsulation.

Outdoor measurements. For the field measurements, two sets of samples were prepared. For temperature tracking of the devices, monolithic perovskite/silicon tandem solar cells (without metal grids), single-junction semitransparent perovskite solar cells (without metal grids) and single-junction SHJ cells were prepared over a 50 cm² area to mimic the realistic applications. All of these devices were encapsulated between two glasses (1 mm low iron glasses) with 10 mm wide butyl rubber edge sealants, which were laminated at 100 °C. The butyl rubber sealants were located 3 cm away from the wafers to avoid additional heating from the sealants. The temperature values were recorded directly from the bottom surface of the wafers using PT100 thermocouples and thermal paste. For power tracking, perovskite/silicon tandem solar cell samples were encapsulated with glass/polyolefin/transparent back sheet from Jolywood (FFC-JW30) structure at 140 °C using Ecolam vacuum laminators. The rest of the active area is shaded by non-transparent aluminium tapes to avoid the contribution of the SHJ bottom cell to the current generation.

Photovoltaic outdoor data were acquired using an $I-V$ tracer from EKO (model MP-160) in November 2019. The $I-V$ characteristics and temperature of multiple samples were probed successively using the multiplexers MI-520 and MI-540, respectively (all from EKO). The power values were tracked with a time interval of 10 min and with a 200 mV s⁻¹ scan speed, identical to laboratory measurements. Current-voltage curves were measured with a scan rate of 200 mV s⁻¹. The electronic equipment was kept inside of a cooled house at 23 °C. Data acquisition was performed with a computer running EKO's MP160 software. The global horizontal irradiance on the plane of the devices (mounted on the same structure as the devices) was measured using the pyranometer MS-802 (EKO). The solar cells were mounted on a structure with a tilt angle of 25° and with south-facing orientation. The structure is located in the outdoor testing field on the KAUST campus, near the village of Thuwal, Saudi Arabia (coordinates 22°30'24.94", 39°11'07.374").

Temperature-dependent bandgap modelling. We followed the modified detailed balance methodology in ref. ⁴² to model the temperature dependence of PCE, which incorporates exponential Urbach band tailing effects into the radiative limit efficiency calculation first established by Shockley and Queisser⁴⁴. Instead of assuming an abrupt, step-function at the band edge, subgap absorption—defined by the E_U of the absorber material—is added, as in ref. ⁴². We apply the same approach to calculate the radiative efficiency limit of perovskite/silicon tandem cells using previously measured temperature-dependent data for the E_U of both materials¹. Furthermore, we measure the E_g of the perovskite under study and use well-established parameterizations for the temperature dependence of the c-Si E_g (ref. ⁴⁵) to establish a temperature-dependent radiative efficiency limit model for the perovskite/silicon tandem cell architecture. As in the original calculation of Shockley and Queisser, we assume ideal absorption above the E_g with no reflection losses. To model the subgap absorption, in addition to the temperature-dependent E_U , we assume a single pass in the perovskite top cell and Lambertian light trapping with ideal rear reflection in the c-Si bottom cell.

Module temperature prediction modelling. The empirical equation⁴⁶ below is used for simulating the device operating temperature T_m .

$$T_m = E \times e^{a+b \times v} + T_{amb}$$

Where T_{amb} is the ambient temperature, E is the Sun irradiance, v is the wind speed, and a and b are empirically determined coefficients. We first use the experimental measured outdoor data in November 2019 at Jeddah to fit the parameters a and b . Using the empirical model with the localized fitted parameters, the device temperature for the complete year of 2016 in Jeddah was simulated based on the environment data from photovoltaic geographical information system⁴⁷.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The datasets generated and analysed during the current study are available within the paper, its Supplementary Information and its Source Data files. Source data are provided with this paper.

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Author contributions

E.A. and T.G.A. conceived the idea and designed the experiments. E.A. developed the perovskite top cells, top electrodes and contact layouts and fabricated the tandem devices. E.A. and T.G.A. performed laboratory-based device characterizations. J.Á. contributed the fabrication of the perovskite absorber. T.G.A. developed nanocrystalline recombination junction and fabricated c-Si bottom cells. M.D.B. performed temperature-dependent absorptance measurements and prepared the silicon wafers. M.D.B. and E.V.K. did the encapsulation of the devices. M.S. performed the field measurements. E.V.K. prepared the field data. L.X. performed the temperature prediction calculations. T.G.A. performed the temperature-dependent radiative efficiency modelling and yield calculations. E.A., T.G.A. and S.D.W. composed the manuscript. All authors discussed the results, contributed to the writing and commented on the manuscript.

Competing interests

The authors declare no competing interests.

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Solar Cells Reporting Summary

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► Experimental design

Please check: are the following details reported in the manuscript?

1. Dimensions

- Area of the tested solar cells ☒ Yes ☐ No 0.8317 cm² aperture area was used. All details are in the certificate (supplementary information) and main text.
- Method used to determine the device area ☒ Yes ☐ No It has been determined by a certification laboratory (CalLab, Fraunhofer ISE)

2. Current-voltage characterization

- Current density-voltage (J-V) plots in both forward and backward direction ☒ Yes ☐ No In the J-V graphs in the main text and supplementary information
- Voltage scan conditions ☒ Yes ☐ No The voltage scan were done both Jsc to Voc and Voc to Jsc. The scan speed is given in the methods section. Dwell time was not applied during the measurements.
For instance: scan direction, speed, dwell times
- Test environment ☒ Yes ☐ No All details about the measurement conditions are added in the methods section.
For instance: characterization temperature, in air or in glove box
- Protocol for preconditioning of the device before its characterization ☒ Yes ☐ No J-V measurements were performed after the cell had undergone maximum-power-point tracking for 300 s. This was done to ensure that the cell reached a stable steady-state efficiency before J-V characterization. Details are added to the methods section.
- Stability of the J-V characteristic ☒ Yes ☐ No Pmpp, Jmpp and Vmpp values were recorded during MPP tracking measurements. In certificate, Fraunhofer ISE CalLab also reported stabilized power.
Verified with time evolution of the maximum power point or with the photocurrent at maximum power point; see [ref. 7](#) for details.

3. Hysteresis or any other unusual behaviour

- Description of the unusual behaviour observed during the characterization ☒ Yes ☐ No The observed hysteresis is negligible and mostly originated from the FF changes. Jmpp and Vmpp values were recorded during MPP tracking measurements to show the stabilized power.
- Related experimental data ☒ Yes ☐ No Fig. 1d and Supplementary Fig. 2

4. Efficiency

- External quantum efficiency (EQE) or incident photons to current efficiency (IPCE) ☒ Yes ☐ No In the main text.
- A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator ☒ Yes ☐ No Fig. 1e
- For tandem solar cells, the bias illumination and bias voltage used for each subcell ☒ Yes ☐ No Reported in the methods section.

5. Calibration

- Light source and reference cell or sensor used for the characterization ☒ Yes ☐ No For in house measurements, AAA class LED based solar simulator was calibrated by using Fraunhofer ISE CalLab certified c-Si cell.

Confirmation that the reference cell was calibrated and certified	☐ Yes ☒ No	The measurement of the device was carried out by the Farunhofer ISE CalLab.
Calculation of spectral mismatch between the reference cell and the devices under test	☐ Yes ☒ No	The measurement of the device was carried out by the Farunhofer ISE CalLab.
6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	In the main text and certificate given in the caption of Supplementary Figure 1.
Variation of the measured short-circuit current density with the mask/aperture area	☐ Yes ☒ No	We report our results always with aperture. Edge effects are minimized. Farunhofer ISE CalLab measured the aperture area by themselves.
7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	CalLab, Fraunhofer ISE.
A copy of any certificate(s) <i>Provide in Supplementary Information</i>	☒ Yes ☐ No	Given in Supplementary Figure 1.
8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Given in Figure S... in the Supplementary Information.
Statistical analysis of the device performance	☒ Yes ☐ No	Given in Supplementary Figure 3.
9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions <i>For instance: illumination type, temperature, atmosphere humidity, encapsulation method, preconditioning temperature</i>	☒ Yes ☐ No	We performed field measurements. The details are given in Fig. 2a. and Fig. 2b.